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Radar-based single target vital sensing

Abstract

In an embodiment, a method includes: generating a target displacement signal indicative of a movement of a human target based on raw digital data generated by a millimeter-wave radar sensor; and estimating a vital sign of the human target based on the target displacement signal, where generating the target displacement signal includes: generating target in-phase (I) and quadrature (Q) (I/Q) data associated with the human target based on the raw digital data, classifying the target I/Q data as high quality data or as low quality data using a first neural network, when the target I/Q data is classified as low quality data, discarding the target I/Q data, when the target I/Q data is classified as high quality data, performing ellipse fitting on the target I/Q data to generate compensated I/Q data, and generating the target displacement signal based on the compensated I/Q data.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is related to co-pending U.S. patent application Ser. No. 16/853,011, filed on Apr. 20, 2020, entitled “Radar-Based Vital Sign Estimation,” and associated with and to co-pending U.S. patent application Ser. No. 16/794,904, filed on Feb. 19, 2020 entitled “Radar Vital Signal Tracking Using a Kalman Filter,” which applications are hereby incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates generally to an electronic system and method, and, in particular embodiments, to a radar-based single target vital sensing.

BACKGROUND

(3) Applications in the millimeter-wave frequency regime have gained significant interest in the past few years due to the rapid advancement in low cost semiconductor technologies, such as silicon germanium (SiGe) and fine geometry complementary metal-oxide semiconductor (CMOS) processes. Availability of high-speed bipolar and metal-oxide semiconductor (MOS) transistors has led to a growing demand for integrated circuits for millimeter-wave applications at e.g., 24 GHz, 60 GHz, 77 GHz, and 80 GHz and also beyond 100 GHz. Such applications include, for example, automotive radar systems and multi-gigabit communication systems.

(4) In some radar systems, the distance between the radar and a target is determined by transmitting a frequency modulated signal, receiving a reflection of the frequency modulated signal (also referred to as the echo), and determining a distance based on a time delay and/or frequency difference between the transmission and reception of the frequency modulated signal. Accordingly, some radar systems include a transmit antenna to transmit the radio-frequency (RF) signal, and a receive antenna to receive the reflected RF signal, as well as the associated RF circuits used to generate the transmitted signal and to receive the RF signal. In some cases, multiple antennas may be used to implement directional beams using phased array techniques. A multiple-input and multiple-output (MIMO) configuration with multiple chipsets can be used to perform coherent and

non-coherent signal processing as well.

SUMMARY

(5) In accordance with an embodiment, a method includes: transmitting radar signals using a millimeter-wave radar sensor; receiving reflected radar signals using the millimeter-wave radar sensor; generating raw digital data based on the reflected radar signals; generating a target displacement signal indicative of a movement of a human target based on the raw digital data; and estimating a vital sign of the human target based on the target displacement signal, where generating the target displacement signal includes: generating target in-phase (I) and quadrature (Q) (I/Q) data associated with the human target based on the raw digital data, classifying the target I/Q data as high quality data or as low quality data using a first neural network, when the target I/Q data is classified as low quality data, discarding the target I/Q data, when the target I/Q data is classified as high quality data, performing ellipse fitting on the target I/Q data to generate compensated I/Q data, and generating the target displacement signal based on the compensated I/Q data.

(6) In accordance with an embodiment, a radar system includes: a millimeter-wave radar sensor including: a transmitter configured to transmit radar signals, a receiver configured to receive reflected radar signals, and an analog-to-digital converter (ADC) configured to generate raw digital data based on the reflected radar signals; and a processing system configured to: generate target in-phase (I) and quadrature (Q) (I/Q) data associated with a human target based on the raw digital data, classify the target I/Q data as high quality data or as low quality data using a first neural network, when the target I/Q data is classified as low quality data, discard the target I/Q data, when the target I/Q data is classified as high quality data, perform ellipse fitting on the target I/Q data to generate compensated I/Q data, generate a target displacement signal indicative of a movement of the human target, and estimate a heartbeat rate of the human target based on the target displacement signal.

(7) In accordance with an embodiment, a method includes: transmitting radar signals using a millimeter-wave radar; receiving reflected radar signals using the millimeter-wave radar; generating raw digital data based on the reflected radar signals; generating a target displacement signal indicative of a movement of a human target based on the raw digital data; performing wavelet denoising on the target displacement signal to generate a denoised displacement signal; using an adaptive Sinc filter to generate a heartbeat filtered displacement signal based on the denoised displacement signal; and estimating a heartbeat rate of the human target based on the heartbeat filtered displacement signal.

(8) In accordance with an embodiment, a method includes: transmitting radar signals using a millimeter-wave radar sensor; receiving reflected radar signals using the millimeter-wave radar sensor; generating raw digital data based on the reflected radar signals; determining a number of people within a field-of-view of the millimeter-wave radar sensor based on the raw digital data; when the number of people is equal to 1, using a vital sensing pipeline to generate a target displacement signal indicative of a movement of a human target based on the raw digital data, and estimate a vital sign of the human target based on the target displacement signal; and when the number of people is greater equal to 0, disabling the vital sensing pipeline.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of the present invention, and the advantages thereof, reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 shows a radar system, according to an embodiment of the present invention;

- (3) FIGS. 2 and 3 show flow charts of embodiment methods for estimating heartbeat rate and respiration rate of a human target, according to an embodiment of the present invention;
- (4) FIG. 4 shows a flow chart of an embodiment method for performing vital sensing on a single human target based on the number of human targets in a field-of-view (FoV) of the millimeter-wave radar sensor of FIG. 1, according to an embodiment of the present invention;
- (5) FIG. 5 shows a flow chart of an embodiment method for generating a displacement signal, according to an embodiment of the present invention;
- (6) FIGS. 6 and 7 show exemplary in-phase (I) and quadrature (Q) plots of low quality I/Q data and high quality I/Q data, respectively;
- (7) FIG. 8 shows a block diagram of an embodiment for estimating heartbeat rate and respiration rate of a single human target from the displacement signal of FIG. 5, according to an embodiment of the present invention;
- (8) FIG. 9 shows an exemplary displacement signal, and identified peaks for estimating a heartbeat rate, according to an embodiment of the present invention;
- (9) FIG. 10 shows a block diagram of a heartbeat rate sensing pipeline, according to an embodiment of the present invention;
- (10) FIGS. 11-13 shows block diagram of embodiments for estimating heartbeat rate and respiration rate of a single human target from the displacement signal of FIG. 5, according to an embodiment of the present invention;
- (11) FIG. 14 shows a block diagram of a heartbeat rate sensing pipeline, according to an embodiment of the present invention;
- (12) FIG. 15 shows a flow chart of an embodiment method for counting people in a scene, according to an embodiment of the present invention;
- (13) FIG. 16 shows a schematic diagram of a possible implementation of the millimeter-wave radar sensor of FIG. 2, according to an embodiment of the present invention;
- (14) FIG. 17 shows a sequence of chirps transmitted by the transmitting antenna of FIG. 16, according to an embodiment of the present invention;
- (15) FIGS. 18 and 19 illustrate heartbeat rate estimations of an embodiment compared with the ground truth, according to an embodiment of the present invention; and
- (16) FIG. 20 shows a device, according to an embodiment of the present invention.
- (17) Corresponding numerals and symbols in different figures generally refer to corresponding parts unless otherwise indicated. The figures are drawn to clearly illustrate the relevant aspects of the preferred embodiments and are not necessarily drawn to scale.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

(18) The making and using of the embodiments disclosed are discussed in detail below. It should be appreciated, however, that the present invention provides many applicable inventive concepts that can be embodied in a wide variety of specific contexts. The specific embodiments discussed are merely illustrative of specific ways to make and use the invention, and do not limit the scope of the invention.

(19) The description below illustrates the various specific details to provide an in-depth understanding of several example embodiments according to the description. The embodiments may be obtained without one or more of the specific details, or with other methods, components, materials and the like. In other cases, known structures, materials or operations are not shown or described in detail so as not to obscure the different aspects of the embodiments. References to “an embodiment” in this description indicate that a particular configuration, structure or feature described in relation to the embodiment is included in at least one embodiment. Consequently, phrases such as “in one embodiment” that may appear at different points of the present description do not necessarily refer exactly to the same embodiment. Furthermore, specific formations, structures or features may be combined in any appropriate manner in one or more embodiments.

(20) Embodiments of the present invention will be described in specific contexts, e.g., a radar-

based system and method for estimating vital signs (such as respiration rate and heartbeat rate) for a single human target. Embodiments of the present invention may be used for estimating vital signs of a non-human target (such as an animal). Some embodiments may be used for post-processing purposes after extracting individual target I/Q data, e.g., in a multi-person scenario.

(21) Embodiments of the present invention may be used in a variety of applications. For example, some embodiments may be used for patient monitoring in hospitals. Some embodiments may be used for sleep monitoring, e.g., at home. Some embodiments may be used for monitoring a user in front of a television or laptop. Some embodiments may be used for monitoring a driver of a car. Other applications are also possible.

(22) In an embodiment of the present invention, a vital sign (e.g., heartbeat rate, respiration rate) of a single human target is estimated (e.g., continuously) using a millimeter-wave radar system. A Good/Bad classifier is used to distinguish between high quality I/Q data and low quality I/Q data associated with the human target. A displacement signal is generated based on the high quality I/Q data, and an adaptive set of Sinc filters filter the displacement signal to generate a filtered displacement signal. The vital sign is then estimated based on the filtered displacement signal. In some embodiments, the vital sensing pipeline that estimates the vital signs of the human target is enabled or disabled based on the number of people present in a field-of-view (FoV) of the millimeter-wave radar system.

(23) A radar, such as a millimeter-wave radar, may be used to detect a human target and track vital signs (such as heartbeat rate and respiration rate) of the detected human target. In some embodiments, therefore, a radar, such as a millimeter-wave radar, enables a contactless, non-invasive method for vital sensing, which may advantageously increase the comfort of the human target during the vital signs monitoring. Some embodiments allow for continuous monitoring of vital signs, which may advantageously allow for detecting critical health conditions in a timely manner.

(24) Conventionally, radar-based vital sensing utilizes the discrete Fourier transform and determines respiration rate and heartbeat rate by detecting the maximum peaks in the frequency spectrum. Such an approach may be susceptible to multiple reflections, multipath effects, motion artifacts, random body movements (RBM) and intermodulation product (IMP), for example. For example, IMP at a frequency of $f_{\text{sub.h}} + f_{\text{sub.r}}$ may occur between the heartbeat rate (e.g., at a heartbeat frequency $f_{\text{sub.h}}$ between 0.7 Hz and 3 Hz) and a respiration rate (e.g., at a respiration frequency $f_{\text{sub.r}}$ between 0.2 Hz and 0.5 Hz).

(25) FIG. 1 shows radar system **100**, according to an embodiment of the present invention. Radar system **100** includes millimeter-wave radar sensor **102** and processing system **104**.

(26) During normal operation, millimeter-wave radar sensor **102** transmits a plurality of radar signals **106**, such as chirps, towards scene **108** using transmitter (TX) circuit **120**. In some embodiments the chirps are linear chirps (i.e., the instantaneous frequency of the chirp varies linearly with time).

(27) The transmitted radar signals **106** are reflected by objects in scene **108**. The reflected radar signals (not shown in FIG. 1), which are also referred to as the echo signal, are received by millimeter-wave radar sensor **102** using receiver (RX) circuit **122** and processed by processing system **104** to, for example, detect a human target and track the vital signs of the detected human target.

(28) The objects in scene **108** may include one or more humans. Other objects, such as furniture and walls, and periodic movement equipment, such as fans, may also be present in scene **108**.

(29) Generally, monitoring vital signs, such as heartbeat rate or respiration rate of a human target, with a radar-based system is a complex endeavor. For example, the amplitude of the heartbeat signal is generally smaller than the amplitude of the respiration signal of the human target. The amplitude of the heartbeat signal is also generally smaller than the amplitude caused by the movement of the human target (e.g., when walking), as well as RBM of the human target (e.g.,

such as lifting an arm, twisting the torso, shaking of the legs, etc.). Additionally, the signal shape of a single heartbeat may be dependent on the subject, the chosen measurement spot, and the distance to the antenna.

(30) FIG. 2 shows a flow chart of embodiment method **200** for estimating heartbeat rate and respiration rate of a human target, according to an embodiment of the present invention. Radar system **100** may implement method **200**. For example, in some embodiments, steps **202** and **204** may be performed by millimeter-wave radar sensor **102** and step **205** (which includes steps **206**, **208**, **210**, **212**, **214**, **216**, **218**, **220**, and **222**) may be performed by processing system **104**.

(31) During step **202**, millimeter-wave radar **102** sensor transmits, e.g., linear chirps organized in frames using transmitter (TX) circuit **120**. The time between chirps of a frame is generally referred to as pulse repetition time (PRT). In some embodiments, the time interval between the end of the last chirp of a frame and the start of the first chirp of the next frame is the same as the PRT so that all chirps are transmitted (and received) equidistantly.

(32) In some embodiments, the chirps have a bandwidth of 2 GHz within the 60 GHz UWB band, the frame time has a duration of 1.28 s, and the PRT is 5 ms (corresponding to an effective sampling rate of 200 Hz). Other values may also be used.

(33) After reflection from objects in scene **108**, receiver (RX) circuit **122** receives reflected radar signals during step **204**. During step **204**, raw digital data (also referred to as raw data or radar data) are generated based on the reflected radar signals received by millimeter-wave radar **102**. For example, in some embodiments, during step **204**, the transmitted and received radar signals are mixed to generate an intermediate frequency (IF) signal. The IF signal may be referred to as the beat signal. The IF signal is then filtered (e.g., with a low-pass and/or band-pass filter) and digitized with an analog-to-digital converter (ADC) to generate the raw data.

(34) During step **205**, one or more vital signs (e.g., S.sub.HBR_f, S.sub.RR_f) of a single human target are estimated based on the raw data. For example, in some embodiments, steps **206**, **208** and **210** are performed to generate a target displacement signal D based on the raw data. The target displacement signal D is then filtered (e.g., during steps **212** and/or **218**) to generate a vital sign filtered displacement signal (e.g., D.sub.f_HBR, D.sub.f_RR). An estimate of the vital sign (e.g., S.sub.HBR, S.sub.RR) is then generated based on the vital sign filtered displacement signal (e.g., during steps **214** and/or **220**). A filtered vital sign (e.g., S.sub.HBR_f, S.sub.RR_f) is then generated based on the estimated vital sign of the human target.

(35) For example, during step **206**, a range fast Fourier transform (FFT) is performed on the raw data to generate range data. For example, in some embodiments, the raw data are zero-padded and the FFT is applied to generate the range data, which may include range information of a plurality of targets.

(36) In some embodiments, the range FFT is applied on all samples of the observation window. The observation window may be implemented as consecutive windows or as sliding windows and may have a length of one or more frames. For example, in some embodiments, the observation window is implemented as a sliding window in which the length of the observation window corresponds to a plurality of time steps that are evaluated during each time step. For example, in an embodiment in which the time step is equal to 1 frame, and the observation window is a sliding window with 8 frames, then, for each frame, the last 8 frames are used as the observation window. In an embodiment, an observation window with a duration of 8 frames has a duration of about 10 s. In some embodiments, each frame includes only a single chirp and the observation window includes 1 chirp. In some embodiments, the observation window is not limited to frames and may include any number of chirps.

(37) In some embodiments, range data, such as a range image, such as a range-Doppler image (RDI) or a range-angle image (RAI) is generated during step **206**.

(38) During step **208**, detection of potential targets is performed. For example, in some embodiments, potential targets are identified by comparing power levels of the range image with a

threshold, where points above the threshold are labeled as targets while points below the threshold are labeled as non-targets. In some embodiments, the range of interest associated with the detected target (the target distance or target bin) is identified based on the location of peaks of the range image having power levels above the threshold.

(39) In some embodiments, an order statistics (OS) constant false alarm rate (CFAR) (OS-CFAR) detector is performed during step to identify potential targets.

(40) In some embodiments, target range bins are clustered to “fuse” the target point cloud belonging to one target to a single target and thus determine the mean range of such single target. For example, in an embodiment, a density-based spatial clustering of applications with noise (DBSCAN) algorithm is used to associate targets to clusters. Other clustering algorithms may also be used. In some embodiments, the clustered targets are used to identify the range of interest associated with the detected target (the target distance or target bin).

(41) The output of step **208** is target I/Q data associated with detected targets.

(42) During step **210**, a target displacement signal D is generated for each detected target based on the respective I/Q data, where the displacement signal D is indicative of the movement of the respective detected target. As will be described in more detail below, in some embodiments, moving targets, and targets exhibiting RBM or IMP may be ignored (e.g., the displacement signal D for such targets may not be generated or may not be further processed, and/or an estimated respiration rate $S_{\text{sub}}\text{RR}$ and estimated heartbeat rate $S_{\text{sub}}\text{HBR}$ may not be generated).

(43) In some embodiments, a single displacement signal D associated with a single target is generated during step **210**. In some embodiments in which multiple targets are detected, a single displacement signal D is generated, where the single displacement signal is associated with the target exhibiting the highest power (e.g., in the radar image).

(44) During step **212**, filtering is performed on the displacement signal D to generate a heartbeat filtered displacement signal $D_{\text{sub}}\text{f_HBR}$ (e.g., of the single target). As will be described in more detail below, in some embodiments, the step of filtering **212** includes performing wavelet denoising and sinc filtering on the displacement signal D to generate the filtered displacement signal $D_{\text{sub}}\text{f_HBR}$. The corner frequencies used during step **212** may be designed/selected to extract heartbeat rate information. For example, in some embodiments, the filtering performed during step **212** may be designed to extract information in the 0.5 Hz to 3.5 Hz, for example. Other ranges are also possible.

(45) During step **214**, the heartbeat rate $S_{\text{sub}}\text{HBR}$ (e.g., of the single target) is estimated based on the filtered displacement signal $D_{\text{sub}}\text{f_HBR}$. As will be described in more detail below, in some embodiments, estimating the heartbeat rate may involve performing an FFT or counting peaks of the filtered displacement signal $D_{\text{sub}}\text{f_HBR}$.

(46) During step **216**, Kalman filtering is performed on the estimated heartbeat rate $S_{\text{sub}}\text{HBR}$ to generate a final heartbeat rate estimate $S_{\text{sub}}\text{HBR}_\text{f}$.

(47) As illustrated in FIG. 2, steps **212**, **214** and **216** are part of heartbeat rate sensing pipeline **203**. As also illustrated in FIG. 2, a similar pipeline (e.g., **209**) may be used for sensing other vital signs, such as respiration rate. For example, during step **218**, filtering is performed on the displacement signal D to generate a respiration filtered displacement signal $D_{\text{sub}}\text{f_RR}$ (e.g., of the single target). In some embodiments, steps **218** may be performed in a similar manner as step **212**, except with corner frequencies designed/selected to extract respiration rate information. For example, in some embodiments, the filtering performed during step **218** may be designed to extract information in the 0.1 Hz to 0.4 Hz, for example. Other ranges are also possible.

(48) During step **220**, the respiration rate $S_{\text{sub}}\text{RR}$ (e.g., of the single target) is estimated based on the filtered displacement signal $D_{\text{sub}}\text{f_RR}$. In some embodiments, steps **220** may be performed in a similar manner as step **214**.

(49) During step **222**, Kalman filtering is performed on the estimated respiration rate $S_{\text{sub}}\text{RR}$ to generate a final respiration rate estimate $S_{\text{sub}}\text{RR}_\text{f}$.

- (50) In some embodiments, the Kalman filters used during steps **216** and **220** may be implemented in any way known in the art.
- (51) In some embodiments, performing Kalman filtering on the heartbeat rate $S_{\text{sub.HBR}}$ and respiration rate $S_{\text{sub.RR}}$ advantageously results in a smoother final heartbeat rate estimate $S_{\text{sub.HBR}_f}$ and respiration rate estimate $S_{\text{sub.RR}_f}$, respectively.
- (52) In some embodiments, steps **216** and/or **222** may be omitted. For example, in some embodiments, step **222** may be omitted to save on computational costs (so that $S_{\text{sub.RR}_f}$ is equal to $S_{\text{sub.RR}}$) while step **216** may be performed to improve the accuracy of the heartbeat rate estimate. Other implementations are also possible.
- (53) In some embodiments, the estimated heartbeat rate $S_{\text{sub.HBR}_f}$ and estimated respiration rate $S_{\text{sub.RR}_f}$ are displayed, e.g., in a screen of a mobile phone or computer. In some embodiments, the estimated heartbeat rate $S_{\text{sub.HBR}_f}$ and estimated respiration rate $S_{\text{sub.RR}_f}$ are transmitted to an external user (e.g., a remote server) for further processing and/or storage. In some embodiments, the estimated heartbeat rate $S_{\text{sub.HBR}_f}$ and estimated respiration rate $S_{\text{sub.RR}_f}$ are stored in volatile or non-volatile memory (e.g., of a mobile device, computer or remote server), e.g., for later retrieval and/or further processing. Other uses of the estimated heartbeat rate $S_{\text{sub.HBR}_f}$ and estimated respiration rate $S_{\text{sub.RR}_f}$ are also possible. For example, in some embodiments, the estimated heartbeat rate $S_{\text{sub.HBR}_f}$ and estimated respiration rate $S_{\text{sub.RR}_f}$ are used for sleep monitoring and analysis, such as for identifying and predicting sleep patterns over a time window.
- (54) In some embodiments, the accuracy of the estimated vital sign(s) (e.g., heartbeat rate or respiration rate) of a single human target may be affected by the number of human targets present in scene **108**. For example, when more than 1 human target is present in scene **108**, the quality/accuracy of the estimated vital sign(s) for the single human target may be (e.g., substantially) degraded.
- (55) In some embodiments, the number of human targets present in scene **108** is counted, and such count is used to enable/disable the vital sensing pipeline (e.g., for performing step **205**) and/or to produce an indication of the confidence level of the vital sign estimate(s) (e.g., $S_{\text{sub.HBR}_f}$ and $S_{\text{sub.RR}_f}$). For example, FIG. 3 shows a flow chart of embodiment method **300** for estimating heartbeat rate and respiration rate of a single human target, according to an embodiment of the present invention. Method **300** includes steps **202**, **204**, **205**, **302** and **304**. Steps **202**, **204** and **205** may be performed in a similar manner as described with respect to method **200**. In some embodiments, processing system **104** performs steps **302** and **304**.
- (56) During step **302**, the number of people CNT in scene **108** is counted. For example, in some embodiments, the number of people CNT in scene **108** is counted in a conventional manner.
- (57) During step **304**, an action is decided based on the number of people CNT in scene **108**. For example, in some embodiments, the vital sign estimation step **205** may be performed or not performed based on the number of people CNT present in scene **108**. For example, in some embodiments, if the number of people CNT is zero, then the vital sign estimation step **205** is not performed (e.g., the vital sensing pipeline for estimating vital signs of the single human target is disabled during step **304**). By not estimating vital signs when no human target is detected in scene **108**, some embodiments advantageously achieve lower power consumption and/or avoid displaying erroneous vital sign estimates.
- (58) In some embodiments, if the number of people CNT is greater than 1 (e.g., 2, 3, or more), a low confidence signal $S_{\text{sub.low_confidence}}$ may be asserted (activated) during step **304** to indicate that the vital estimates (e.g., $S_{\text{sub.HBR}_f}$, $S_{\text{sub.RR}_f}$) of the single human target may be inaccurate (e.g., since the presence of a one or more additional human targets in scene **108** may interfere with the vital sign estimates). In some embodiments, if the number of people CNT is greater than 1, then the vital signal estimation step **205** is not performed (e.g., the vital sensing pipeline for estimating vital signs of the single human target is disabled during step **304**).

(59) In some embodiments, if the number of people CNT is equal to 1, step 205 is performed (e.g., vital sensing pipeline is/remains enabled during step 304 so that vital sign estimates (e.g., S.sub.HBR_f, S.sub.RR_f) are generated during step 205), and the low confidence signal S.sub.low_confidence may be deasserted (deactivated) during step 304 to indicate that the vital estimates (e.g., S.sub.HBR_f, S.sub.RR_f) of the single human target are not degraded by the presence of additional human targets in scene 108.

(60) FIG. 4 shows a flow chart of embodiment method 400 for performing vital sensing on a single human target based on the number of human targets in a field-of-view (FoV) of millimeter-wave radar sensor 102, according to an embodiment of the present invention. Step 304 may be implemented as method 400.

(61) During step 402, the number of people CNT in scene 108 (e.g., in the FoV of millimeter-wave radar sensor 102) is received. If the number of people CNT is zero (i.e., no human target is detected in scene 108), then the vital sensing pipeline (e.g., 205) is disabled during step 404.

(62) If it is determined during step 402 that there is at least one human target in scene 108 (output “no” from step 402), then if CNT is equal to 1 (i.e., if there is only one human target detected in scene 108), then the vital sensing pipeline is enabled or kept enabled during step 408 so that vital signs are estimated (e.g., during step 205).

(63) If it is determined during step 406 that there is more than one human target in scene 108 (output “no” from step 406), then the range of the main target (e.g., the target associated with the vital sign estimates performed or to be performed during step 205, such as the human target associated with displacement signal D) is compared with a predetermined threshold R.sub.thres. If the main target is farther than the predetermined threshold R.sub.thres, then the vital sensing pipeline is disabled during step 404. If the main target is closer than the predetermined threshold R.sub.thres, then the low confidence signal S.sub.low_confidence is asserted during step 412, and the vital sensing pipeline is or is kept enabled during step 404.

(64) In some embodiments, the accuracy of vital sign estimates of a main target in a multitarget scene is higher in short range settings (output “no” from step 410) than in long range settings (output “yes” from step 410). Thus, in some embodiments, by disabling the vital sensing pipeline in a multi-target long range vital sensing setting, some embodiments advantageously save power and computational costs and avoid determining and/or providing vital sign estimates with low accuracy. In some embodiments, by providing vital signs of the main target in a multi-target short range vital sensing settings while asserting the low confidence signal, some embodiments advantageously provide estimates of the vital signs of the main target while indicating that the confidence in such estimates is low.

(65) Examples of short range settings include: a human target in front of a laptop at a distance lower than R.sub.thres (where the laptop includes millimeter-wave radar sensor 102), a human target driving a car at a distance lower than R.sub.thres from the millimeter-wave radar sensor 102 (which may be located, e.g., in a steering wheel or dashboard of a car), and a human target sleeping next to a mobile phone at a distance lower than R.sub.thres (where the mobile phone includes millimeter-wave radar sensor 102). Examples of long range settings include: a human target sitting in a sofa in front of a television at a distance higher than R.sub.thres (where the television includes millimeter-wave radar sensor 102), and a human target located (e.g., sitting, standing) at a distance higher than R.sub.thres from a security camera (where the security camera includes millimeter-wave radar sensor 102).

(66) In some embodiments, R.sub.thres may be, e.g., about 1.2 meters. Other values (e.g., higher than 1.2 meters, such as 1.5 meters or higher, or lower than 1.2 meters, such as 1 meter, or lower) may also be used.

(67) FIG. 5 shows a flow chart of embodiment method 500 for generating a displacement signal, according to an embodiment of the present invention. Step 210 may be performed as method 500.

(68) During step 502, target I/Q data S.sub.I/Q_i associated with observation windows i for a (e.g.,

single) detected target is high-passed filtered. In some embodiments, the observation window covers one or more frames. In some embodiments, a corner frequency of the high-pass filter used during step **502** is 20 Hz. Other frequencies, such as higher than 20 Hz (e.g., 23 Hz, Hz or higher), or lower than 20 Hz (e.g., 18 Hz, 10 Hz, or lower) may also be used.

(69) During step **504**, the power associated with target I/Q data $S_{\text{sub.I/Q}_i}$ is estimated, e.g., in a known manner, to generate power estimate $P_{\text{sub.HP}}$. If power estimate $P_{\text{sub.HP}}$ is higher than a predetermined threshold $P_{\text{sub.HP_thres}}$ (output “yes” from step **506**), then the target I/Q data $S_{\text{sub.I/Q}_i}$ is discarded during step **508**, and the next observation window is processed (as shown by step **528**). By discarding target I/Q data associated with high power at high frequencies (e.g., high power at frequencies higher than the corner frequency of the high-pass filter used during step **502**), some embodiments advantageously discard and avoid processing target I/Q data that may be unsuitable for extracting vital signs. For example, in some embodiments, performing vital sign estimation involves estimating micro-Doppler signals below, e.g., 5 Hz. Movement of the human target, such as walking or running, may prevent the extraction of accurate vital signs (e.g., $S_{\text{sub.HBR}_f}$, $S_{\text{sub.RR}_f}$) of a detected human target. Thus, in some embodiments, the high-frequency power $P_{\text{sub.HP}}$ serves as an indication of the amount of movement of the human target during observation window i . Thus, by discarding samples associated with high power at high frequencies, some embodiments advantageously avoid processing target I/Q data that may be unsuitable for estimating accurate vital signs (e.g., $S_{\text{sub.HBR}_f}$, $S_{\text{sub.RR}_f}$).

(70) If power estimate $P_{\text{sub.HP}}$ is not higher than a predetermined threshold $P_{\text{sub.HP_thres}}$ (output “no” from step **506**), then the target I/Q data $S_{\text{sub.I/Q}_i}$ is further processed during step **510**. During steps **510** and **512**, target I/Q data $S_{\text{sub.I/Q}_i}$ is low-pass filtered and down-sampled to generate filtered I/Q data $S_{\text{sub.I/Q_filt}_i}$. By low-pass filtering the target I/Q data $S_{\text{sub.I/Q}_i}$, some embodiments advantageously remove high frequency noise, which may aid in improving the accuracy of the vital sign estimates (e.g., $S_{\text{sub.HBR}_f}$, $S_{\text{sub.RR}_f}$).

(71) In some embodiments, a corner frequency of the low-pass filter used during step **510** may be 20 Hz. Other frequencies, such as higher than 20 Hz (e.g., 23 Hz, 25 Hz or higher), or lower than 20 Hz (e.g., 18 Hz, 10 Hz, or lower) may also be used.

(72) In some embodiments, the corner frequency of the low-pass filter used during step **510** may be equal to the corner frequency of the high-pass filter used during step **502**. In some embodiments, the corner frequency of the low-pass filter used during step **510** and the high-pass filter used during step **502** may be different.

(73) In some embodiments, using a corner frequency for the low-pass filter and high-pass filter at or around 20 Hz (e.g., between 15 Hz and 25 Hz) advantageously avoids filtering out information that may be useful in estimating vital signs while removing high frequency noise (e.g., related to large body movements). In some embodiments, any small body movement retained in filtered I/Q data $S_{\text{sub.I/Q_filt}_i}$ may be addressed (e.g., compensated for), e.g., in steps **514** and **522**.

(74) During step **514**, the filtered I/Q data $S_{\text{sub.I/Q_filt}_i}$ is classified using a neural network. For example, in some embodiments, a deep neural network (DNN) performs a multilevel classification to classify the filtered I/Q data $S_{\text{sub.I/Q_filt}_i}$ into 3 classes: RBM (**516**), IMP (**518**), and Good Data (**520**). Classes RBM (**516**) and IMP (**518**) correspond to low quality data while class Good Data (**520**) corresponds to high quality data. Exemplary I-Q plots of low quality I/Q data and high quality I/Q data are shown in FIGS. **6** and **7**, respectively.

(75) In some embodiments, the neural network used during step **514** may be implemented as a SincNet plus CNN layers in a known manner. Other implementations are also possible.

(76) In some embodiments, the neural network (e.g., SincNet plus CNN layers) used during step **514** may be trained in a known manner, e.g., using supervised training.

(77) As shown in FIG. **5**, filtered I/Q data $S_{\text{sub.I/Q_filt}_i}$ classified as RBM or IMP is discarded during step **508** (and is not further processed), and the next observation window is processed (as shown by step **528**). By discarding filtered I/Q data $S_{\text{sub.I/Q_filt}_i}$ classified as RBM or IMP,

some embodiments advantageously discard and avoid processing I/Q data that may be unsuitable for extracting vital signs.

(78) If the filtered I/Q data $S_{\text{sub.I/Q_filt_i}}$ is classified during step **514** as Good Data (**520**), then ellipse fitting is performed in a known manner on the filtered I/Q data $S_{\text{sub.I/Q_filt_i}}$ during step **522** to generate compensated I/Q data $S_{\text{sub.I/Q_comp_i}}$. For example, during step **522**, a conventional ellipse fitting algorithm (also referred to as ellipse correction algorithm) is applied to the I-Q trace (of the complex range data) associated with the detected target to compensate for offset, amplitude, and gain errors. In some embodiments, the compensated I/Q data $S_{\text{sub.I/Q_comp_i}}$ are I-Q signals that correspond to the best fit ellipse associated with the filtered I/Q data $S_{\text{sub.I/Q_filt_i}}$. Some embodiments may avoid using the ellipse fitting algorithm. For example, some embodiments may use an offset compensation algorithm during step **522** or may avoid implementing step **522**. In some embodiments, a neural network may be used for generating the compensated I/Q data $S_{\text{sub.I/Q_comp_i}}$ based on the filtered I/Q data $S_{\text{sub.I/Q_filt_i}}$. Other implementations are also possible.

(79) During step **524**, the angle of the compensated I/Q data $S_{\text{sub.I/Q_comp_i}}$ is calculated by arctangent demodulation of the I-Q signals from the selected range bin selected during step **208** (the I-Q signals associated with the detected target). The resulting phase values in the range of $[-\pi, +\pi]$ are unwrapped between two consecutive data points during step **526**. For example, during step **526**, the phase is unwrapped by adding or subtracting 2π for phase jumps larger than $-\pi$ or $+\pi$, respectively, to keep the phase between $-\pi$ or $+\pi$.

(80) In some embodiments, steps **524** and **526** may be performed by calculating the displacement signal D as

$$(81) \quad D = \frac{\lambda}{4\pi} \cdot \text{Math.unwrap}(\arctan \frac{Q}{I}) \quad (1)$$

where D represents the time-domain displacement signal, λ is the wavelength of the carrier frequency of the chirps (e.g., transmitted during step **202**), $\lambda/2$ represents the unambiguousness (phase) range, and I and Q are the in-phase and quadrature-phase components of the carrier, respectively, associated with the detected target.

(82) As shown in FIG. 5, in some embodiments, once the displacement signal D is determined for an observation window i , the next observation window $i++$ is processed.

(83) In some embodiments, a module performing the steps **502**, **504**, **506**, **508**, **510**, **512**, **514**, **516**, **618**, and **520**, is referred to as a Good/Bad classifier **501**. As illustrated in FIG. 5, the Good/Bad classifier **501** provides filtered I/Q data to be processed during step **522** containing I/Q data classified as good data (**520**) while not providing I/Q data classified as RBM (**516**) or IMP (**518**) as well as not providing I/Q data of human target exhibiting large movements (e.g., step **506**, output “yes”).

(84) FIG. 8 shows a block diagram of embodiment module **800** for estimating heartbeat rate and respiration rate of a single human target from target displacement signal D , according to an embodiment of the present invention. Module **800** includes wavelet denoiser **802**, heartbeat rate sensing pipeline **803** and respiration rate sensing pipeline **809**. Heartbeat rate sensing pipeline **803** includes adaptive Sinc filter **804**, heartbeat rate estimator **806** and Kalman filter **808**. Respiration rate sensing pipeline **809** includes adaptive Sinc filter **810**, respiration rate estimator **812** and Kalman filter **814**. Processing system **104** may implement module **800**. Heartbeat rate sensing pipeline **203** may be implemented as heartbeat rate sensing pipeline **803**. Respiration rate sensing pipeline **209** may be implemented as respiration rate sensing pipeline **809**.

(85) As shown in FIG. 8, wavelet denoiser **802** receives target displacement signal D (e.g., from step **526**) and generates denoised displacement signal $D_{\text{sub.f}}$. For example, in some embodiments, wavelet denoiser **802** performs a wavelet transform on target displacement signal D in a known manner, removes wavelet coefficients that fall below a threshold, and then perform an inverse wavelet transform on the resulting coefficients to generate the denoised displacement signal

D.sub.f.

(86) As shown in FIG. 8, adaptive Sinc filter **804** receives denoised displacement signal D.sub.f and generates heartbeat filtered displacement signal D.sub.f_HBR, where the settings of the adaptive Sinc filter are based on one or more selection signals S.sub.sinc_HBR. As will be described in more detail below, selection signal(s) S.sub.sinc_HBR may come from Kalman filter **808**, may be based on signal S.sub.RR, or may be generated by another block, such as a prediction block.

(87) In some embodiments, adaptive Sinc filter **804** is configured to change the filter settings (e.g., change the corner frequencies of the Sinc filters) to facilitate estimation of the heartbeat rate by heartbeat rate estimator **806**.

(88) Heartbeat rate estimator **806** is configured to estimate the heartbeat rate S.sub.HBR from heartbeat filtered displacement signal D.sub.f_HBR. In some embodiments, heartbeat rate estimator **806** estimates the heartbeat rate S.sub.HBR by counting the number of peaks in heartbeat filtered displacement signal D.sub.f_HBR and estimating the heartbeat rate S.sub.HBR based on the number of detected peaks per unit of time. For example, FIG. 9 shows an exemplary displacement signal D.sub.f_HBR, and identified peaks for estimating heartbeat rate S.sub.HBR, according to an embodiment of the present invention. In some embodiments, heartbeat rate estimator **806** estimates the heartbeat rate S.sub.HBR by performing an FFT on displacement signal D.sub.f_HBR and detecting the highest peak of the FFT (where the heartbeat rate S.sub.HBR is estimated as the frequency corresponding to the location of the peak of the FFT). Other implementations of heartbeat rate estimator **806**, such as based on measuring the time between peaks in heartbeat filtered displacement signal D.sub.f_HBR, is also possible.

(89) Kalman filter **808** is configured to provide a prediction of the heartbeat rate S.sub.HBR_f based on the history of previous heartbeat rates and on the current estimated heartbeat rate S.sub.HBR. In some embodiments, Kalman filter **808** advantageously smoothens the heartbeat rate estimation. Kalman filter **808** may be implemented in any way known in the art.

(90) In some embodiments, modules **810**, **812**, and **814** may be implemented in an analogous manner as modules **804**, **806**, and **808**, but targeting the respiration rate as opposed to the heartbeat rate. For example, the corner frequencies of adaptive Sinc filter **810** for generating respiration filtered displacement signal D.sub.f_RR may be different (e.g., lower) than the corner frequencies of adaptive Sinc filter **810**.

(91) As shown in FIG. 8, in some embodiments, steps **212** and **218** may be implemented with Sinc filters **804** and **810**, respectively, and with a shared wavelet denoiser **802**.

(92) In some embodiments, one or more modules **804**, **808**, **810**, and **814** may be omitted. In some embodiments, only heartbeat rate is estimated (and modules **810**, **812**, and **814**) may be omitted. In some embodiments, only respiration rate is estimated (and modules **804**, **806**, and **808** may be omitted). Other implementations are also possible.

(93) FIG. 10 shows a block diagram of heartbeat rate sensing pipeline **1000**, according to an embodiment of the present invention. Heartbeat rate sensing pipeline **1000** includes adaptive Sinc filter **1002**, heartbeat rate estimator **1004**, and Kalman filter **808**. Adaptive Sinc filter **1002** includes Sinc filter bank **1006** and selector **1008**. Heartbeat rate estimator **1004** includes peak counter **1010**, FFT module **1012**, SNR determination module **1014**, and selector **1016**. Heartbeat rate sensing pipeline **803** may be implemented as heartbeat rate sensing pipeline **1000**. Adaptive Sinc filter **804** may be implemented as adaptive Sinc filter **1002**. Heartbeat rate estimator **806** may be implemented as heartbeat rate estimator **1004**.

(94) In some embodiments, peak counter **1010** is configured to count peaks in a displacement signal (e.g., D.sub.f_HBR), e.g., as illustrated in FIG. 9, and extract a frequency (e.g., in beats per minute) based on the number of peaks detected per unit time. In some embodiments, peak counter **1010** may be implemented in any way known in the art.

(95) In some embodiments, FFT module **1012** is configured to perform an FFT on a displacement

signal (e.g., D.sub.f_HBR) to generate a frequency spectrum and extract a frequency (e.g., in beats per minute) based on the location of a (e.g., highest) peak in the frequency spectrum. In some embodiments, FFT module **1012** may be implemented in any way known in the art.

(96) In some embodiments, SNR module **1014** is configured to extract the SNR of a displacement signal (e.g., D.sub.f_HBR). SNR module **1014** may be implemented in any way known in the art.

(97) Sinc filter bank **1006** is configured to receive denoised displacement signal D.sub.f (e.g., from wavelet denoiser **802**) and apply M Sinc filters to denoised displacement signal D.sub.f to generate M corresponding outputs. In some embodiments, the corner frequencies of each of the M Sinc filters of Sinc filter bank **1006** are fixed.

(98) In some embodiments, the bandwidth of each Sinc filter of Sinc filter bank **1006** is the same. For example, in some embodiments, the bandwidth of each Sinc filter of Sinc filter bank **1006** is 0.1 Hz, f.sub.1 is 0.7 Hz, and f.sub.M is 3 Hz (so that M is 24). Other values for f.sub.1, f.sub.M, and the bandwidth of each Sinc filter of Sinc filter bank **1006** may also be used. For example, in some embodiments, the bandwidth of each Sinc filter of Sinc filter bank **1006** may be higher than 0.1 Hz (e.g., 0.15 Hz, 0.2 Hz, or higher) or lower than 0.1 Hz (e.g., 0.08 Hz, 0.05 Hz, or lower). In some embodiments, corner frequency f.sub.1 may be lower than 0.7 Hz (e.g., 0.65 Hz, 0.6 Hz or lower), or higher than 0.7 Hz (e.g., 0.8 Hz, 0.9 Hz or higher). In some embodiments, corner frequency f.sub.M may be lower than 3 Hz (e.g., 2.9 Hz, 2.7 Hz or lower), or higher than 3 Hz (3.2 Hz, 3.5 Hz or higher). In some embodiments, M may be lower than 24 (e.g., 20, 16, 12, 8, or lower), or higher than 24 (e.g., 28, 32, or higher). In some embodiments, M may be greater than 1.

(99) In some embodiments, the bandwidth of each Sinc filter of Sinc filter bank **1006** may be different (e.g., broader or more granular).

(100) Selector module **1008** is configured to select one or more outputs from the M outputs of Sinc filter bank **1006** based on final heartbeat rate estimate S.sub.HBR_f and heartbeat rate variance S.sub.HBR_var (where variance S.sub.HBR_var represents the variance of the (e.g., recent) history of final heartbeat rate estimates S.sub.HBR_f) and produce displacement signal D.sub.f_HBR, where displacement signal D.sub.f_HBR is formed by concatenating each of the selected outputs of Sinc filter bank **1006**. For example, when the variance S.sub.HBR_var is very low (e.g., a person with a stable heartbeat rate, such as varying less than 2 bpm across a relatively long time window, such as 1 minute, e.g., near the heartbeat rate initialization value of Kalman filter **216**), selector **1008** may select a single output of Sinc filter bank **1006** based on the final heartbeat rate estimate S.sub.HBR_f. For example, in an embodiment in which each Sinc filter of Sinc filter bank **1006** has the same bandwidth of 0.1 Hz, if final heartbeat rate estimate S.sub.HBR_f is 80 bpm (1.33 Hz) and the variance S.sub.HBR_var is very low, then selector **1008** selects the output of the Sinc filter from 1.3 Hz to 1.4 Hz as the displacement signal D.sub.f_HBR.

(101) Some embodiments may, instead of using variance S.sub.HBR_var for selecting the number of Sinc filters to select from Sinc filter bank **1006**, use a fixed number of Sinc filters. For example, in some embodiments, if final heartbeat rate estimate S.sub.HBR_f is 80 bpm (1.33 Hz), selector module **1008** may select two adjacent Sinc filters above (e.g., the Sinc filter from 1.3 Hz to 1.4 Hz and the Sinc filter from 1.4 Hz to 1.5 Hz) and three adjacent Sinc filters below (e.g., the Sinc filter from 1.2 Hz to 1.3 Hz, the Sinc filter from 1.1 Hz to 1.2 Hz, and the Sinc filter from 1 Hz to 1.1 Hz) irrespective of the value of variance S.sub.HBR_var.

(102) In some embodiments, if the final heartbeat rate estimate S.sub.HBR_f is at a boundary between two adjacent Sinc filters and the variance S.sub.HBR_var is very low, then selector **1008** may select the two adjacent outputs and concatenate them to produce displacement signal D.sub.f_HBR. For example, in an embodiment in which each Sinc filter of Sinc filter bank **1006** has the same bandwidth of 0.1 Hz, if final heartbeat rate estimate S.sub.HBR_f is 78 bpm (1.3 Hz) and the variance S.sub.HBR_var is very low, then selector **1008** selects the output of the Sinc filter from 1.2 Hz to 1.3 Hz and the output of the Sinc filter from 1.3 Hz to 1.4 Hz, and concatenate both outputs to form the displacement signal D.sub.f_HBR.

(103) In some embodiments, as the variance $S_{\text{sub.HBR_var}}$ increases, selector **1008** selects more outputs from Sinc filter bank **1006** (based on how high variance $S_{\text{sub.HBR_var}}$ is). When variance $S_{\text{sub.HBR_var}}$ is very high (e.g., when the heartbeat rate jumps from 80 bpm to 120 bpm in a relatively short time window, such as within 10 seconds), selector **1008** may select all M outputs of Sinc filter bank **1006** and concatenate them to form the displacement signal $D_{\text{sub.f_HBR}}$.

(104) As shown in FIG. **10**, heartbeat rate estimator **1004** may estimate heartbeat rate $S_{\text{sub.HBR}}$ using peak counter **1010** or FFT **1012** based on the signal-to-noise ratio (SNR) of displacement signal $D_{\text{sub.f_HBR}}$. For example, in some embodiments, SNR module **1014** determines the SNR ($\text{SNR}_{\text{sub.DHBR}}$) of displacement signal $D_{\text{sub.f_HBR}}$. If $\text{SNR}_{\text{sub.DHBR}}$ is higher than a predetermined SNR threshold ($\text{SNR}_{\text{sub.thres}}$), heartbeat rate estimator **1004** estimate heartbeat rate $S_{\text{sub.HBR}}$ using an FFT (selector **1016** selects the output of FFT **1012** and provides it to Kalman filter **808**). If $\text{SNR}_{\text{sub.DHBR}}$ is lower than the predetermined SNR threshold ($\text{SNR}_{\text{sub.thres}}$) heartbeat rate estimator **1004** estimate heartbeat rate $S_{\text{sub.HBR}}$ by counting peaks of displacement signal $D_{\text{sub.f_HBR}}$ (selector **1016** selects the output of peak counter **1919** and provides it to Kalman filter **808**).

(105) A non-limiting example of a situation exhibiting a high SNR higher than SN threshold $\text{SNR}_{\text{sub.thres}}$ in some embodiments is when a person sits in a very stable manner in front of the millimeter-wave radar sensor **102** in an otherwise empty room.

(106) In some embodiments, during the beginning of normal operation (e.g., when the vital sensing pipeline used during step **205** is enabled), selector **1008** initially selects all M outputs of Sinc filter bank **1006**. As Kalman filter **808** generates predictions for heartbeat rate $S_{\text{sub.HBR_f}}$ and variance $S_{\text{sub.HBR_var}}$ decreases, selector **1008** begins reducing the number of selected outputs from Sinc filter bank **1006**.

(107) In some embodiments, respiration rate sensing pipeline **809** may be implemented in an analogous manner as heartbeat rate sensing pipeline **809** (e.g., with corner frequencies suitable for extracting respiration rate from a displacement signal).

(108) In some embodiments, the respiration rate of a human target is much more stable than the heartbeat rate of the human target. Thus, in some embodiments, the complexity of the respiration rate estimator may be lower than the complexity of the heartbeat rate estimator while maintaining comparable accuracy. For example, FIG. **11** shows a block diagram of embodiment module **1100** for estimating heartbeat rate and respiration rate of a single human target from displacement signal D , according to an embodiment of the present invention. Module **1100** includes heartbeat rate sensing pipeline **1000**, respiration rate sensing pipeline **1102**, and wavelet denoiser **802**. Respiration rate sensing pipeline **809** may be implemented as respiration rate sensing pipeline **1102**. Respiration rate estimator **812** may be implemented as peak counter **1106**. Peak counter **1106** may be implemented in a similar manner as peak counter **1010**. Processing system **104** may implement module **1100**.

(109) As shown in FIG. **11**, in some embodiments, respiration rate sensing pipeline **1102** may be implemented with a Sinc filter **1104** for generating filtered displacement signal $D_{\text{sub.f_RR}}$, and peak counter **1106** for generating final respiration rate estimate $S_{\text{sub.RR_f}}$. In some embodiments, Sinc filter **1104** may be implemented with a low corner frequency $f_{\text{sub.1}}$ equal to 0.2 Hz and a high corner frequency $f_{\text{sub.h}}$ equal to 0.6 Hz. Other corner frequency values, such as with $f_{\text{sub.1}}$ lower than 0.2 Hz (e.g., 0.15 Hz, 0.1 Hz, or lower) or higher than 0.2 Hz (e.g., 0.25 Hz or higher) and such as with $f_{\text{sub.h}}$ lower than 0.6 Hz (e.g., 0.55 Hz, 0.5 Hz, or lower) or higher than 0.6 Hz (e.g., 0.65 Hz or higher).

(110) The inventors realized that it is common for the respiration rate and the heartbeat rate to be correlated. For example, it is common for the heartbeat rate to be on the lower end of the heartbeat rate range when the respiration rate is on the lower end of the respiration rate range. Similarly, it is common for the heartbeat rate to be on the higher end of the heartbeat rate range when the respiration rate is on the higher end of the respiration rate range.

(111) The inventors also realized that harmonics of the respiration rate (e.g., the third harmonics) may be in the range of a possible heartbeat rate and thus may interfere with heartbeat rate estimation. For example, the third harmonic of a respiration rate of 16 bpm is at 0.81 Hz. Thus, the third harmonics of a respiration rate of 16 bpm may result in additional peaks in the heartbeat displacement signal $D_{\text{sub.f_HBR}}$, which may cause an estimation of the heartbeat rate that is higher than the actual heartbeat rate.

(112) Thus, in some embodiments, the settings of the adaptive Sinc filter are based on the respiration rate $S_{\text{sub.RR_f}}$. For example, FIG. 12 shows a block diagram of embodiment module 1200 for estimating heartbeat rate and respiration rate of a single human target from displacement signal D , according to an embodiment of the present invention. Module 1200 includes heartbeat rate sensing pipeline 1201, respiration rate sensing pipeline 1102, and wavelet denoiser 802. Heartbeat rate sensing pipeline 1201 includes adaptive Sinc filter 1202 and heartbeat rate estimator 1004. Adaptive Sinc filter 1202 includes Sinc filter bank 1006 and selector 1208. Heartbeat rate sensing pipeline 803 may be implemented as heartbeat rate sensing pipeline 1200. Adaptive Sinc filter 804 may be implemented as adaptive Sinc filter 1202. Processing system 104 may implement module 1200.

(113) In some embodiments, the set of outputs available to selector 1208 is based on the respiration rate $S_{\text{sub.RR_f}}$. For example, in some embodiments, if the respiration rate $S_{\text{sub.RR_f}}$ is lower than a predetermined respiration rate threshold $RR_{\text{sub.thres}}$, then a low set of outputs of Sinc filter bank 1006 (e.g., from $f_{\text{sub.1}}$ to $f_{\text{sub.Q}}$, Q being greater than 1 and lower than M) is available to selector 1208 to select outputs in a similar manner as selector 1008. If the respiration rate $S_{\text{sub.RR_f}}$ is higher than the predetermined respiration rate threshold $RR_{\text{sub.thres}}$, then a high set of outputs of Sinc filter bank 1006 (e.g., from $f_{\text{sub.P}}$ to $f_{\text{sub.M}}$, P being greater than 1 and lower than M , and preferably lower than Q) is available to selector 1208 to select outputs in a similar manner as selector 1008. For example in an embodiment in which each Sinc filter of Sinc filter bank 1006 has the same bandwidth of 0.1 Hz, $f_{\text{sub.1}}$ is equal to 0.7 Hz and $f_{\text{sub.M}}$ is equal to 2 Hz, Q may be 1.4 Hz and P may be 0.9 Hz, and $RR_{\text{sub.thres}}$ may be 16 bpm (0.27 Hz). In such embodiment, when the respiration rate $S_{\text{sub.RR_f}}$ is below 16 bpm, only outputs between 0.7 Hz and 1.4 Hz (0.7 Hz to 0.8 Hz, 0.8 Hz to 0.9 Hz, . . . , 1.2 Hz to 1.3 Hz, and 1.3 Hz to 1.4 Hz) are available for selector 1208 to choose from and when the respiration rate $S_{\text{sub.RR_f}}$ is above 16 bpm, only outputs between 0.9 Hz and 2 Hz (0.9 Hz to 1 Hz, 1 Hz to 1.1 Hz, . . . , 1.8 Hz to 1.9 Hz, 1.9 Hz to 2 Hz) are available for selector 1208 to choose from.

(114) In some embodiments, $RR_{\text{sub.thres}}$ may be higher than 16 bpm (such as 17 bpm or higher) or lower than 16 bpm (such as 15 bpm or lower). In some embodiments, $RR_{\text{sub.thres}}$ is not fixed.

(115) In some embodiments, it is possible that the respiration rate be low (e.g., below $RR_{\text{sub.thres}}$) and that the heartbeat rate be outside the bandwidth of the low set of outputs of Sinc filter bank 1006 (e.g., higher than $f_{\text{sub.Q}}$). Thus, in some embodiments, the selection between the low set of outputs and the high set of outputs may be overridden, e.g., based on the SNR $SNR_{\text{sub.DHBR}}$ of the displacement signal $D_{\text{sub.f_HBR}}$. For example, FIG. 13 shows a block diagram of embodiment module 1300 for estimating heartbeat rate and respiration rate of a single human target from displacement signal D , according to an embodiment of the present invention. Module 1300 includes heartbeat rate sensing pipeline 1301, respiration rate sensing pipeline 1102, and wavelet denoiser 802. Heartbeat rate sensing pipeline 1301 includes adaptive Sinc filter 1302 and heartbeat rate estimator 1004. Adaptive Sinc filter 1302 includes Sinc filter bank 1006 and selector 1308. Heartbeat rate sensing pipeline 803 may be implemented as heartbeat rate sensing pipeline 1300. Adaptive Sinc filter 804 may be implemented as adaptive Sinc filter 1302. Processing system 104 may implement module 1300.

(116) As shown in FIG. 13, selector 1308 may select one or more outputs of Sinc filter bank 1006 based on heartbeat rate $S_{\text{sub.HBR_f}}$, respiration rate $S_{\text{sub.RR_f}}$, variance $S_{\text{sub.HBR_var}}$, and SNR $SNR_{\text{sub.DHBR}}$. For example, if the SNR $SNR_{\text{sub.DHBR}}$ (e.g., of the previous estimate) is

higher than a predetermined SNR threshold (e.g., SNR.sub.thres), the respiration rate S.sub.RR_f is below RR.sub.thres, and the heartbeat rate (e.g., of the previous estimate) is higher than f.sub.Q, then selector **1308** selects from the high set of outputs of Sinc filter bank **1006** one or more outputs in a similar manner as selector **1008** (in contrast to selecting from the low set of outputs of Sinc filter bank **1006** if using, e.g., module **1200**). For example, in an embodiment in which each Sinc filter of Sinc filter bank **1006** has the same bandwidth of 0.1 Hz, f.sub.1 is equal to 0.7 Hz, f.sub.M is equal to 2 Hz, Q is 1.4 Hz, P is 0.9 Hz, and RR.sub.thres is 16 bpm (0.27 Hz), if the respiration rate S.sub.RR_f is below 16 bpm, the heartbeat rate S.sub.HBR_f is above 1.4 Hz, and SNR.sub.DHBR is above SNR.sub.thres, then selector **1308** selects from the 12 available outputs of the high set (from 0.9 Hz to 2 Hz) in a similar manner as selector **1008** (e.g., based on S.sub.HBR_var and S.sub.HBR_f).

(117) As shown in FIGS. **10-13**, the adaptive filter (e.g., **1002**, **1102**, **1202**, **1302**) may change its settings based on one or more outputs of Kalman filter **808**. Thus, in some embodiments, the adaptive filter (e.g., **1002**, **1102**, **1202**, **1302**) may change its settings in a retrospective manner (based on the history of previous heartbeat rate estimates S.sub.HBR_f). Some embodiments may change the settings of the adaptive filter in a prospective manner (e.g., based on the current sample D.sub.f and without regard for the previous heartbeat rate estimates S.sub.HBR_f). For example, FIG. **14** shows a block diagram of heartbeat rate sensing pipeline **1400**, according to an embodiment of the present invention. Heartbeat rate sensing pipeline **1400** includes heartbeat rate range predictor **1404**, adaptive Sinc filter **1402**, heartbeat rate estimator **1004**, and Kalman filter **808**. Adaptive Sinc filter **1402** includes Sinc filter bank **1006** and selector **1408**. Heartbeat rate sensing pipeline **803** may be implemented as heartbeat rate sensing pipeline **1400**. Adaptive Sinc filter **804** may be implemented as adaptive Sinc filter **1402**. Processing system **104** may implement heartbeat rate sensing pipeline **1400**.

(118) Heartbeat rate range predictor **1404** is configured to predict the frequency range of the heartbeat rate information of the human target based on displacement signal D.sub.f. For example, in some embodiments, heartbeat rate range predictor **1404** determines in which range of the M ranges of the M outputs of Sinc filter bank **1006** the heartbeat rate of the human target is based on the displacement signal D.sub.f and provides such frequency range (HBR.sub.pred) to selector **1402**. Selector **1408** then selects one or more outputs of the M outputs of Sinc filter bank **1006** based on the frequency range HBR.sub.pred. For example, in some embodiments, selector **1408** selects a single output of Sinc filter bank **1006** that corresponds to the frequency range HBR.sub.pred. In some embodiments, selector **1408** selects more than one output of Sinc filter bank **1006**. For example, in some embodiments, selector **1408** selects three outputs of Sinc filter bank **1006**, e.g., the output that corresponds to the frequency range HBR.sub.pred and the two adjacent outputs (the frequency range above frequency range HBR.sub.pred and the frequency range below frequency range HBR.sub.pred). In some embodiments, the frequency range HBR.sub.pred corresponds to a plurality of outputs of Sinc filter bank **1006**. Other implementations are also possible.

(119) In some embodiments, heartbeat rate range predictor **1404** is implemented as a classifier with M classes (a class for each of the M ranges of the M outputs of Sinc filter bank **1006**). For example, in some embodiments, heartbeat rate range predictor **1404** is implemented as a deep neural network (DNN). The DNN may be trained to predict the frequency ranges of the heartbeat rate based on displacement signal D.sub.f. In some embodiments, training the DNN to predict ranges may be advantageously less complex (e.g., may be done in substantially less iterations) than predicting exact values of the heartbeat rate.

(120) As a non-limiting example, in an embodiment in which each Sinc filter of Sinc filter bank **1006** has the same bandwidth of 0.1 Hz, f.sub.1 is equal to 0.7 Hz, and f.sub.M is equal to 2 Hz, the DNN of heartbeat rate range predictor **1404** classifies displacement signal D.sub.f as one of 14 possible classes by determining the frequency range HBR.sub.pred of the heartbeat rate out of 14

possible frequency ranges (0.7 Hz to 0.8 Hz, 0.8 Hz to 0.9 Hz, . . . , 1.8 Hz to 1.9 Hz, and 1.9 Hz to 2 Hz).

(121) In some embodiments, frequency range HBR.sub.pred corresponds to a plurality of outputs (e.g., 2, 3, or more) of Sinc filter bank **1006**. Other implementations are also possible.

(122) FIG. **15** shows a flow chart of embodiment method **1500** for counting people in scene **108**, according to an embodiment of the present invention. Step **302** may be performed as method **1500**.

(123) During step **1502**, raw data is received (e.g., from millimeter-wave radar sensor **102**) and RDIs are generated from the received raw data, e.g., in a known manner. For example, in some embodiments, frame integration is performed on the received raw data and MTI filtering is performed on the integrated frames. Then, a range FFT is applied to the integrated and filtered frames using a sliding window and a Doppler FFT is applied on the other corresponding axis to generate a plurality of sequential RDIs. Other implementations are also possible.

(124) During step **1504**, a DNN is used to classify the RDIs into 3 possible classes: 0 humans (**1506**), 1 human (**1508**), and more than 1 human (**1510**).

(125) In some embodiments, the DNN used in step **1504** may be trained in a known manner, e.g., using supervised training. By classifying the number of people into 3 classes (**1506**, **1508**, and **1510**) instead of counting the exact number of people in scene **108**, some embodiments may advantageously train the DNN faster and with less iteration than if determining the exact number of people in scene **108**.

(126) In some embodiments, the DNN used in step **1504** may be implemented as a convolutional neural network (CNN) or as a CNN with long short-term memory (LSTM). Other implementations are also possible.

(127) FIG. **16** shows a schematic diagram of a possible implementation of millimeter-wave radar sensor **102**, according to an embodiment of the present invention. In some embodiments, millimeter-wave radar sensor **102** may be used to generate radar raw data (x.sub.out_dig(n)) and provide such raw data to processing system **104** for determining vital signs (e.g., S.sub.HBR_f, S.sub.RR_f) of a human target.

(128) During normal operation, millimeter-wave radar sensor **102** may operate as a frequency-modulated continuous-wave (FMCW) radar sensor and transmits a plurality of TX radar signals **106**, such as chirps, towards scene **108** using one or more transmitter (TX) antenna **1614**. The radar signals **106** are generated using RF and analog circuits **1630**. The radar signals **106** may be, e.g., in the 20 GHz to 122 GHz range. Other frequencies may also be used.

(129) The radar signals **106** are reflected by objects in scene **108**. The reflected radar signals **107**, which are also referred to as the echo signal, are received by one or more receiving (RX) antennas **1616**. RF and analog circuits **1630** processes the received reflected radar signals **107** using, e.g., band-pass filters (BPFs), low-pass filters (LPFs), mixers, low-noise amplifier (LNA), and/or intermediate frequency (IF) amplifiers in ways known in the art to generate an analog signal x.sub.out(t).

(130) The analog signal x.sub.out(t) is converted to raw digital data x.sub.out_dig(n) (also referred to as raw radar data or raw data) using ADC **1612**. The raw digital data x.sub.out_dig(n) is processed by processing system **104** to, e.g., to determine the vital signs (e.g., S.sub.HBR_f, S.sub.RR_f) of a human target in scene **108**.

(131) Controller **1610** controls one or more circuits of millimeter-wave radar sensor **102**, such as RF and analog circuit **1630** and/or ADC **1612**. Controller **1610** may be implemented, e.g., as a custom digital or mixed signal circuit, for example. Controller **1610** may also be implemented in other ways, such as using a general purpose processor or controller, for example. In some embodiments, processing system **104** implements a portion or all of controller **1610**.

(132) Processing system **104** may be implemented with a general purpose processor, controller or digital signal processor (DSP) that includes, for example, combinatorial circuits coupled to a memory and configured to execute instructions stored in the memory. In some embodiments,

processing system **1204** may be implemented as an application specific integrated circuit (ASIC). In some embodiments, processing system **104** may be implemented with an ARM, RISC, or x86 architecture, for example. In some embodiments, processing system **104** may include an artificial intelligence (AI) accelerator. Some embodiments may use a combination of hardware accelerator and software running on a DSP or general purpose microcontroller. In some embodiments, processing system **104** may be implemented with a plurality of processors and/or controllers. Other implementations are also possible.

(133) In some embodiments, millimeter-wave radar sensor **102** and a portion or all of processing system **104** may be implemented inside the same integrated circuit (IC). For example, in some embodiments, millimeter-wave radar sensor **102** and a portion or all of processing system **104** may be implemented in respective semiconductor substrates that are integrated in the same package. In other embodiments, millimeter-wave radar sensor **102** and a portion or all of processing system **104** may be implemented in the same monolithic semiconductor substrate. In some embodiments, millimeter-wave radar sensor **102** and processing system **104** are implemented in respective integrated circuits. In some embodiments, a plurality of integrated circuits is used to implement millimeter-wave radar sensor **102**. In some embodiments, a plurality of integrated circuits is used to implement processing system **104**. Other implementations are also possible.

(134) As a non-limiting example, RF and analog circuits **1630** may be implemented, e.g., as shown in FIG. **16**. During normal operation, voltage-controlled oscillator (VCO) **1636** generates radar signals, such as a linear frequency chirps (e.g., from 57 GHz to 64 GHz, or from 76 GHz to 77 GHz), which are transmitted by transmitting antenna **1614**. The VCO **1636** is controlled by PLL **1634**, which receives a reference clock signal (e.g., 80 MHz) from reference oscillator **1632**. PLL **1634** is controlled by a loop that includes frequency divider **1638** and amplifier **1640**. Amplifier **1637** may be used to drive transmitting antenna **1614**.

(135) The TX radar signals **106** transmitted by one or more transmitting antennas **1614** are reflected by objects in scene **108** and received by one or more receiving antennas **1616**. The echo received by receiving antenna **1616** is mixed with a replica of the signal transmitted by transmitting antenna **1614** using mixer **1646** to produce intermediate frequency (IF) signal $x_{\text{sub}}.\text{IF}(t)$ (also known as beat signal). In some embodiments, the beat signal $x_{\text{sub}}.\text{IF}(t)$ has a bandwidth between 10 kHz and 1 MHz. Beat signals with a bandwidth lower than 10 kHz or higher than 1 MHz is also possible. Amplifier **1645** may be used to receive the reflected radar signals from antenna **1616**.

(136) Beat signal $x_{\text{sub}}.\text{IF}(t)$ is filtered with low-pass filters (LPF) **1648** and then sampled by ADC **1612**. ADC **1212** is advantageously capable of sampling the filtered beat signal $x_{\text{sub}}.\text{out}(t)$ with a sampling frequency that is much smaller than the frequency of the signal received by receiving antenna **1616**. Using FMCW radars, therefore, advantageously allows for a compact and low cost implementation of ADC **1612**, in some embodiments.

(137) The raw digital data $x_{\text{sub}}.\text{out_dig}(n)$, which in some embodiments include the digitized version of the filtered beat signal $x_{\text{sub}}.\text{out}(t)$ is (e.g., temporarily) stored, e.g., in matrices of $N_{\text{sub.c}} \times N_{\text{sub.s}}$ per receiving antenna **1616**, where $N_{\text{sub.c}}$ is the number of chirps considered in a frame and $N_{\text{sub.s}}$ is the number of transmit samples per chirp, for further processing by processing system **104**.

(138) FIG. **17** shows a sequence of chirps **106** transmitted by TX antenna **1614**, according to an embodiment of the present invention. As shown by FIG. **17**, chirps **106** are organized in a plurality of frames and may be implemented as up-chirps. Some embodiments may use down-chirps or a combination of up-chirps and down-chirps, such as up-down chirps and down-up chirps. Other waveform shapes may also be used.

(139) As shown in FIG. **17**, each frame may include a plurality of chirps **106**. For example, in some embodiments, the number of chirps in a frame is 16. Some embodiments may include more than 16 chirps per frame, such as 20 chirps, 32 chirps, or more, or less than 16 chirps per frame, such as 10 chirps, 8 chirps, or less. In some embodiments, each frame includes only a single chirp.

(140) In some embodiments, frames are repeated every FT time. In some embodiments, FT time is 50 ms. A different FT time may also be used, such as more than 50 ms, such as 60 ms, 100 ms, 200 ms, or more, or less than 50 ms, such as 45 ms, 40 ms, or less.

(141) In some embodiments, the FT time is selected such that the time between the beginning of the last chirp of frame n and the beginning of the first chirp of frame n+1 is equal to PRT. Other embodiments may use or result in a different timing.

(142) The time between chirps of a frame is generally referred to as pulse repetition time (PRT). In some embodiments, the PRT is 5 ms. A different PRT may also be used, such as less than 5 ms, such as 4 ms, 2 ms, or less, or more than 5 ms, such as 6 ms, or more.

(143) The duration of the chirp (from start to finish) is generally referred to as chirp time (CT). In some embodiments, the chirp time may be, e.g., 64 μ s. Higher chirp times, such as 128 μ s, or higher, may also be used. Lower chirp times, may also be used.

(144) In some embodiments, the chirp bandwidth may be, e.g., 4 GHz. Higher bandwidth, such as 6 GHz or higher, or lower bandwidth, such as 2 GHz, 1 GHz, or lower, may also be possible.

(145) In some embodiments, the sampling frequency of millimeter-wave radar sensor **902** may be, e.g., 1 MHz. Higher sampling frequencies, such as 2 MHz or higher, or lower sampling frequencies, such as 500 kHz or lower, may also be possible.

(146) In some embodiments, the number of samples used to generate a chirp may be, e.g., 64 samples. A higher number of samples, such as 128 samples, or higher, or a lower number of samples, such as 32 samples or lower, may also be used.

(147) FIGS. **18** and **19** illustrate vital sign outputs of an embodiment compared with the ground truth, according to an embodiment of the present invention. FIG. **18** corresponds to a quasi-static human target while FIG. **19** corresponds to a human target shaking legs and exhibiting RBM.

(148) Curves **1802** and **1902** were generated by performing method **500** to generate the displacement signal D and using heartbeat rate sensing pipeline **1000** to generate the heartbeat rate estimate, with a single human target in front of millimeter-wave radar sensor **102**.

(149) Curves **1804** and **1904** were generated using a conventional heartbeat rate sensor attached the human target. Curves **1804** and **1904** serve as ground truth.

(150) Curves **1806** and **1906** were generated using a millimeter-wave radar in a conventional manner.

(151) As shown in FIG. **18**, curve **1802** is closer to the ground truth (**1804**) than using a conventional method (curve **1806**). In particular, the RMSE score for curve **1802** is lower (2.9143) than for curve **1806** (5.5877).

(152) Similarly, as shown in FIG. **19**, curve **1902** is closer to the ground truth (**1904**) than using a conventional method (curve **1906**). In particular, the RMSE score for curve **1902** is lower (15.8430) than for curve **1906** (187457).

(153) FIG. **20** shows device **2000**, according to an embodiment of the present invention. Device **2000** includes radar system **100**, and screen **2002**.

(154) In some embodiments, radar system **100** estimates vital signs (e.g., S.sub.HBR_f, S.sub.RR_f) of a human sitting in front of screen **2002**, and screen **2002** displays such estimated vital signs.

(155) As shown in FIG. **20**, device **2000** may be implemented as a laptop. In some embodiments, device **2000** may be implemented, e.g., as a mobile phone, television, etc.

(156) Example embodiments of the present invention are summarized here. Other embodiments can also be understood from the entirety of the specification and the claims filed herein.

(157) Example 1. A method including: transmitting radar signals using a millimeter-wave radar sensor; receiving reflected radar signals using the millimeter-wave radar sensor; generating raw digital data based on the reflected radar signals; generating a target displacement signal indicative of a movement of a human target based on the raw digital data; and estimating a vital sign of the human target based on the target displacement signal, where generating the target displacement

signal includes: generating target in-phase (I) and quadrature (Q) (I/Q) data associated with the human target based on the raw digital data, classifying the target I/Q data as high quality data or as low quality data using a first neural network, when the target I/Q data is classified as low quality data, discarding the target I/Q data, when the target I/Q data is classified as high quality data, performing ellipse fitting on the target I/Q data to generate compensated I/Q data, and generating the target displacement signal based on the compensated I/Q data.

(158) Example 2. The method of example 1, where generating the target I/Q data includes: generating preliminary IQ data based on the raw digital data; high-pass filtering the preliminary IQ data to generate a high-pass filtered IQ data using a high-pass cutoff frequency; estimating a power of the high-pass filtered IQ data; when the estimated power is higher than a power threshold, discarding the preliminary IQ data; and when the estimated power is lower than the power threshold, low-pass filtering the preliminary IQ data using a low-pass cutoff frequency to generate the target IQ data.

(159) Example 3. The method of one of examples 1 or 2, where the high-pass cutoff frequency is equal to the low-pass cutoff frequency.

(160) Example 4. The method of one of examples 1 to 3, where classifying the target I/Q data as low quality data includes classifying the target I/Q data as random body movement (RBM) data or as intermodulation product (IMP) data.

(161) Example 5. The method of one of examples 1 to 4, further including performing wavelet denoising on the target displacement signal to generate a denoised displacement signal, where estimating the vital sign of the human target is based on the denoised displacement signal.

(162) Example 6. The method of one of examples 1 to 5, further including performing adaptive Sinc filtering to generate a vital sign filtered displacement signal based on the target displacement signal, where estimating the vital sign of the human target is based on the vital sign filtered displacement signal.

(163) Example 7. The method of one of examples 1 to 6, where performing adaptive Sinc filtering includes generating M Sinc filter outputs using M Sinc filters based on the target displacement signal and generating the vital sign filtered displacement signal based on one or more of the M Sinc filter outputs, where M is a positive integer greater than 1.

(164) Example 8. The method of one of examples 1 to 7, further including filtering the estimated vital sign using a Kalman filter to generate a filtered vital sign.

(165) Example 9. The method of one of examples 1 to 8, further including generating, with the Kalman filter, a vital sign variance associated with the filtered vital sign, where generating the vital sign filtered displacement signal is further based on the filtered vital sign and vital sign variance.

(166) Example 10. The method of one of examples 1 to 9, where the estimated vital sign of the human target is an estimated heartbeat rate of the human target, the method further including estimating a respiration rate of the human target based on the target displacement signal.

(167) Example 11. The method of one of examples 1 to 10, where generating the vital sign filtered displacement signal is further based on the respiration rate.

(168) Example 12. The method of one of examples 1 to 11, where generating the vital sign filtered displacement signal is based on a first sub-set of outputs of the M Sinc filter outputs when the respiration rate is below a respiration rate threshold, and is based on a second sub-set of outputs of the M Sinc filter outputs when the respiration rate is above the respiration rate threshold, the first sub-set of outputs being different from the second sub-set of outputs.

(169) Example 13. The method of one of examples 1 to 12, where the first sub-set of outputs corresponds to outputs of first Sinc filters having adjacent bandwidth, where the second sub-set of outputs corresponds to outputs of second Sinc filters having adjacent bandwidth, and where a low-corner frequency of the collective bandwidth of the first Sinc filters is smaller than a low-corner frequency of the collective bandwidth of the second Sinc filters.

(170) Example 14. The method of one of examples 1 to 13, further including: determining a signal-

to-noise (SNR) ratio of the vital sign filtered displacement signal; when the SNR ratio is above an SNR threshold, performing a Fourier transform on the vital sign filtered displacement signal to estimate the vital sign of the human target; and when the SNR ratio is below the SNR threshold, performing peak counting on the vital sign filtered displacement signal to estimate the vital sign of the human target, where generating the vital sign filtered displacement signal is further based on the SNR ratio.

(171) Example 15. The method of one of examples 1 to 14, further including determining a frequency range of the vital sign of the human target based on the target displacement signal using a deep neural network (DNN), where generating the vital sign filtered displacement signal is further based on the frequency range.

(172) Example 16. The method of one of examples 1 to 15, further including generating the vital sign filtered displacement signal based on a plurality of Sinc filter outputs of the M Sinc filter outputs by concatenating the plurality of Sinc filter outputs.

(173) Example 17. The method of one of examples 1 to 16, where each of the M Sinc filters have the same bandwidth.

(174) Example 18. The method of one of examples 1 to 17, where each of the M Sinc filters have fixed corner frequencies.

(175) Example 19. The method of one of examples 1 to 18, further including displaying the estimated vital sign on a screen.

(176) Example 20. The method of one of examples 1 to 19, where a vital sensing pipeline is used for generating the target displacement signal and estimating the vital sign of the human target, the method further including: determining a number of people within a field-of-view of the millimeter-wave radar sensor; when the number of people is equal to 0, disabling the vital sensing pipeline; and when the number of people is equal to 1, enabling the vital sensing pipeline.

(177) Example 21. The method of one of examples 1 to 20, further including: determining a range of the human target; when the human target is closer than a predetermined range and the number of people is higher than 1, asserting a low confidence signal indicative of low confidence in the estimated vital sign and enabling the vital sensing pipeline; and when the human target is closer than the predetermined range and the number of people is higher than 1, disabling the vital sensing pipeline.

(178) Example 22. The method of one of examples 1 to 21, where the first neural network is a SincNet neural network.

(179) Example 23. A radar system including: a millimeter-wave radar sensor including: a transmitter configured to transmit radar signals, a receiver configured to receive reflected radar signals, and an analog-to-digital converter (ADC) configured to generate raw digital data based on the reflected radar signals; and a processing system configured to: generate target in-phase (I) and quadrature (Q) (I/Q) data associated with a human target based on the raw digital data, classify the target I/Q data as high quality data or as low quality data using a first neural network, when the target I/Q data is classified as low quality data, discard the target I/Q data, when the target I/Q data is classified as high quality data, perform ellipse fitting on the target I/Q data to generate compensated I/Q data, generate a target displacement signal indicative of a movement of the human target, and estimate a heartbeat rate of the human target based on the target displacement signal.

(180) Example 24. A method including: transmitting radar signals using a millimeter-wave radar; receiving reflected radar signals using the millimeter-wave radar; generating raw digital data based on the reflected radar signals; generating a target displacement signal indicative of a movement of a human target based on the raw digital data; performing wavelet denoising on the target displacement signal to generate a denoised displacement signal; using an adaptive Sinc filter to generate a heartbeat filtered displacement signal based on the denoised displacement signal; and estimating a heartbeat rate of the human target based on the heartbeat filtered displacement signal.

(181) Example 25. A method including: transmitting radar signals using a millimeter-wave radar sensor; receiving reflected radar signals using the millimeter-wave radar sensor; generating raw digital data based on the reflected radar signals; determining a number of people within a field-of-view of the millimeter-wave radar sensor based on the raw digital data; when the number of people is equal to 1, using a vital sensing pipeline to generate a target displacement signal indicative of a movement of a human target based on the raw digital data, and estimate a vital sign of the human target based on the target displacement signal; and when the number of people is greater equal to 0, disabling the vital sensing pipeline.

(182) Example 26. The method of example 25, further including, when the number of people is equal to 1, determining a target range associated with the human target; when the target range is higher than a range threshold, disabling the vital sensing pipeline; and when the target range is lower than the range threshold, using the vital sensing pipeline to generate the target displacement signal and estimate a vital sign of the human target based on the target displacement signal, and asserting a low confidence signal indicative of low confidence in the estimated vital sign.

(183) While this invention has been described with reference to illustrative embodiments, this description is not intended to be construed in a limiting sense. Various modifications and combinations of the illustrative embodiments, as well as other embodiments of the invention, will be apparent to persons skilled in the art upon reference to the description. It is therefore intended that the appended claims encompass any such modifications or embodiments.

Claims

1. A method comprising: transmitting radar signals using a millimeter-wave radar sensor; receiving reflected radar signals using the millimeter-wave radar sensor; generating raw digital data based on the reflected radar signals; generating a target displacement signal indicative of a movement of a human target based on the raw digital data; and estimating a vital sign of the human target based on the target displacement signal, wherein generating the target displacement signal comprises: generating target in-phase (I) and quadrature (Q) (I/Q) data associated with the human target based on the raw digital data, classifying the target I/Q data as high quality data or as low quality data using a first neural network, when the target I/Q data is classified as low quality data, discarding the target I/Q data, when the target I/Q data is classified as high quality data, performing ellipse fitting on the target I/Q data to generate compensated I/Q data, and generating the target displacement signal based on the compensated I/Q data; and performing adaptive Sinc filtering to generate a vital sign filtered displacement signal based on the target displacement signal, wherein estimating the vital sign of the human target is based on the vital sign filtered displacement signal.
2. The method of claim 1, wherein generating the target I/Q data comprises: generating preliminary I/Q data based on the raw digital data; high-pass filtering the preliminary I/Q data to generate a high-pass filtered I/Q data using a high-pass cutoff frequency; estimating a power of the high-pass filtered I/Q data; when the estimated power is higher than a power threshold, discarding the preliminary I/Q data; and when the estimated power is lower than the power threshold, low-pass filtering the preliminary I/Q data using a low-pass cutoff frequency to generate the target I/Q data.
3. The method of claim 2, wherein the high-pass cutoff frequency is equal to the low-pass cutoff frequency.
4. The method of claim 1, wherein classifying the target I/Q data as low quality data comprises classifying the target I/Q data as random body movement (RBM) data or as intermodulation product (IMP) data.
5. The method of claim 1, further comprising performing wavelet denoising on the target displacement signal to generate a denoised displacement signal, wherein estimating the vital sign of the human target is based on the denoised displacement signal.
6. The method of claim 1, wherein performing adaptive Sinc filtering comprises generating M Sinc

filter outputs using M Sinc filters based on the target displacement signal and generating the vital sign filtered displacement signal based on one or more of the M Sinc filter outputs, wherein M is a positive integer greater than 1.

7. The method of claim 6, further comprising filtering the estimated vital sign using a Kalman filter to generate a filtered vital sign.

8. The method of claim 7, further comprising generating, with the Kalman filter, a vital sign variance associated with the filtered vital sign, wherein generating the vital sign filtered displacement signal is further based on the filtered vital sign and the vital sign variance.

9. The method of claim 6, wherein the estimated vital sign of the human target is an estimated heartbeat rate of the human target, the method further comprising estimating a respiration rate of the human target based on the target displacement signal.

10. The method of claim 9, wherein generating the vital sign filtered displacement signal is further based on the respiration rate.

11. The method of claim 10, wherein generating the vital sign filtered displacement signal is based on a first sub-set of outputs of the M Sinc filter outputs when the respiration rate is below a respiration rate threshold, and is based on a second sub-set of outputs of the M Sinc filter outputs when the respiration rate is above the respiration rate threshold, the first sub-set of outputs being different from the second sub-set of outputs.

12. The method of claim 11, wherein the first sub-set of outputs corresponds to outputs of first Sinc filters having adjacent bandwidth, wherein the second sub-set of outputs corresponds to outputs of second Sinc filters having adjacent bandwidth, and wherein a low-corner frequency of a collective bandwidth of the first Sinc filters is smaller than a low-corner frequency of a collective bandwidth of the second Sinc filters.

13. The method of claim 10, further comprising: determining a signal-to-noise ratio (SNR) of the vital sign filtered displacement signal; when the SNR is above an SNR threshold, performing a Fourier transform on the vital sign filtered displacement signal to estimate the vital sign of the human target; and when the SNR is below the SNR threshold, performing peak counting on the vital sign filtered displacement signal to estimate the vital sign of the human target, wherein generating the vital sign filtered displacement signal is further based on the SNR.

14. The method of claim 6, further comprising determining a frequency range of the vital sign of the human target based on the target displacement signal using a deep neural network (DNN), wherein generating the vital sign filtered displacement signal is further based on the frequency range.

15. The method of claim 6, further comprising generating the vital sign filtered displacement signal based on a plurality of Sinc filter outputs of the M Sinc filter outputs by concatenating the plurality of Sinc filter outputs.

16. The method of claim 6, wherein each of the M Sinc filters has the same bandwidth.

17. The method of claim 6, wherein each of the M Sinc filters has a fixed corner frequency.

18. The method of claim 1, further comprising displaying the estimated vital sign on a screen.

19. The method of claim 1, wherein a vital sensing pipeline is used for generating the target displacement signal and estimating the vital sign of the human target, the method further comprising: determining a number of people within a field-of-view of the millimeter-wave radar sensor; when the number of people is equal to 0, disabling the vital sensing pipeline; and when the number of people is equal to 1, enabling the vital sensing pipeline.

20. The method of claim 19, further comprising: determining a range of the human target; when the human target is closer than a predetermined range and the number of people is higher than 1, asserting a low confidence signal indicative of low confidence in the estimated vital sign and enabling the vital sensing pipeline; and when the human target is higher than the predetermined range and the number of people is higher than 1, disabling the vital sensing pipeline.

21. The method of claim 1, wherein the first neural network is a SincNet neural network.

22. A radar system comprising: a millimeter-wave radar sensor comprising: a transmitter configured to transmit radar signals, a receiver configured to receive reflected radar signals, and an analog-to-digital converter (ADC) configured to generate raw digital data based on the reflected radar signals; and a processing system configured to: generate target in-phase (I) and quadrature (Q) (I/Q) data associated with a human target based on the raw digital data, classify the target I/Q data as high quality data or as low quality data using a first neural network, when the target I/Q data is classified as low quality data, discard the target I/Q data, when the target I/Q data is classified as high quality data, perform ellipse fitting on the target I/Q data to generate compensated I/Q data, generate a target displacement signal indicative of a movement of the human target based on the compensated I/Q data, estimate a vital sign of the human target based on the target displacement signal, and perform adaptive Sinc filtering to generate a vital sign filtered displacement signal based on the target displacement signal, wherein estimating the vital sign of the human target is based on the vital sign filtered displacement signal.

23. A method comprising: transmitting radar signals using a millimeter-wave radar sensor; receiving reflected radar signals using the millimeter-wave radar sensor; generating raw digital data based on the reflected radar signals; generating a target displacement signal indicative of a movement of a human target based on the raw digital data; and estimating a vital sign of the human target based on the target displacement signal, wherein generating the target displacement signal comprises: generating target in-phase (I) and quadrature (Q) (I/Q) data associated with the human target based on the raw digital data, generating the target I/Q data comprising: generating preliminary I/Q data based on the raw digital data, high-pass filtering the preliminary I/Q data to generate a high-pass filtered I/Q data using a high-pass cutoff frequency, estimating a power of the high-pass filtered I/Q data, when the estimated power is higher than a power threshold, discarding the preliminary I/Q data, and when the estimated power is lower than the power threshold, low-pass filtering the preliminary I/Q data using a low-pass cutoff frequency to generate the target I/Q data, classifying the target I/Q data as high quality data or as low quality data using a first neural network, when the target I/Q data is classified as low quality data, discarding the target I/Q data, when the target I/Q data is classified as high quality data, performing ellipse fitting on the target I/Q data to generate compensated I/Q data, and generating the target displacement signal based on the compensated I/Q data.

24. The method of claim 23, wherein the high-pass cutoff frequency is equal to the low-pass cutoff frequency.
